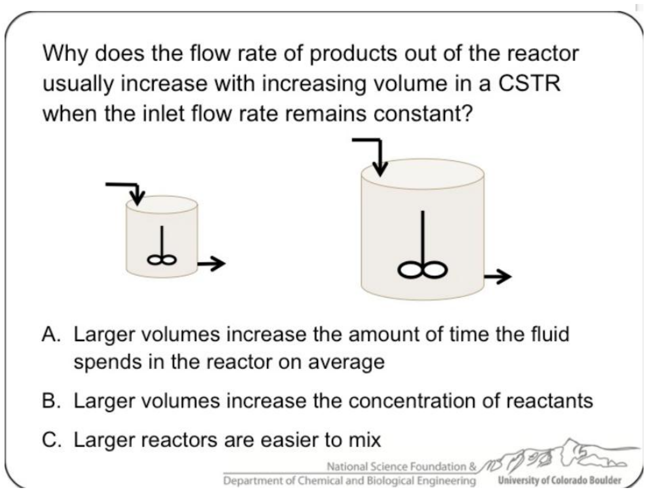
321 Problem Set 1

Liv Lyons

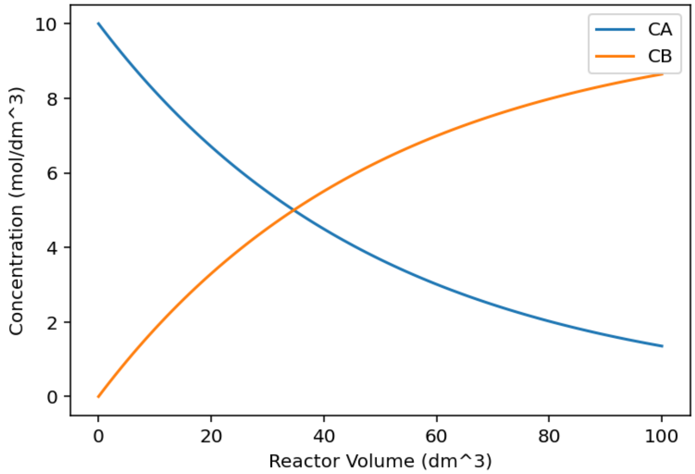
Q1-2)



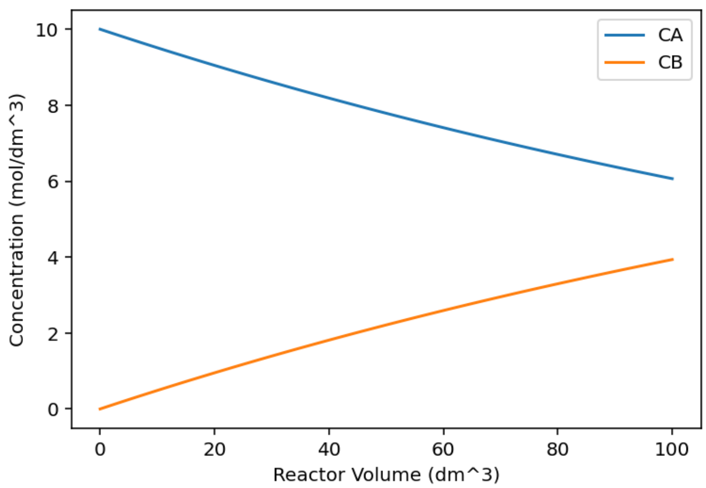
This question would be a good one to have on a test. I feel like as a conceptual, it has a lot of areas to think about. Originally, I thought that it was B so that it would stay in equilibrium, but after some more thought, I realized the answer was A. This is because if the fluid stays in the reactor for longer (larger volume), it has more time to react so more product is produced.

Q1-5) The content goals of the course are to, most importantly, teach chemical reaction engineering clearly by focusing on core principles, real-world applications issues such as reactor safety. The text has fundamental building blocks like mole balances and reactor models. This text, along with the accompanying website, allow students to analyze a wide range of reaction engineering problems through logic and reasoning. As far as intellectual goals go, these focus on critical and creative thinking. The website supports these goals by being a second teacher, engaging and an extension of the course. It also offers simulations using Python and Wolfram, LEPs, and additional notes/materials. When used with the text, the website can help deepen the understanding of the course both content and intellectual goals wise.

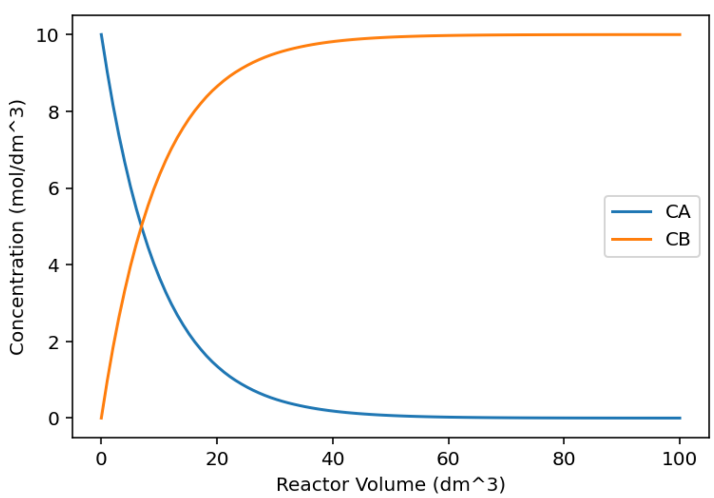
P1-1) 1)



This is the normal graph with k=0.23 and v0=10.



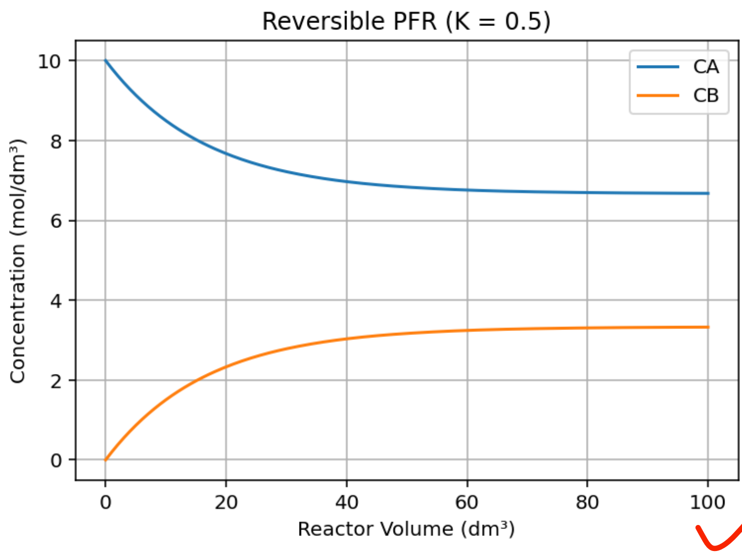
After keeping k at 0.23 and heavily increasing v0 (v0=20), the graph ended up as follows. There is shorter residence time and less conversion. CA stays higher while CB stays lower.

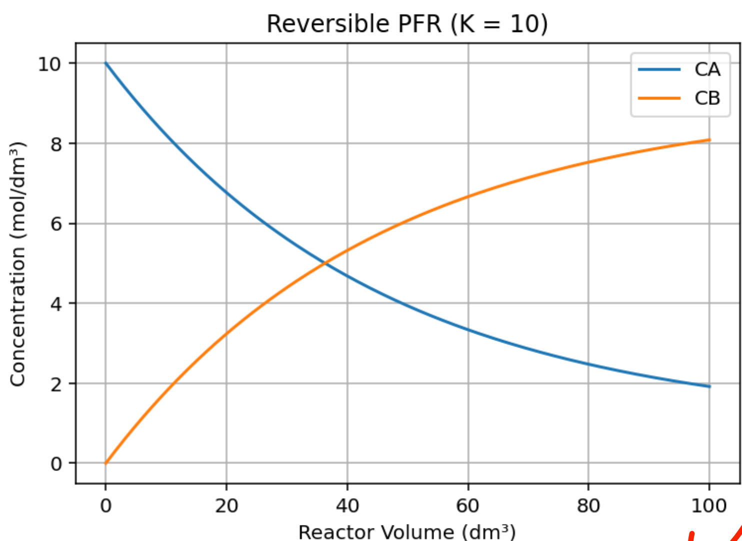


Now, k has been increased to 0.5 and v0 stays at its original value. The reaction is now much faster with CA decreasing faster and CB increasing faster. Conversion increases for the same reactor volume.

Increasing the specific reaction rate constant k causes species A to be consumed quicker along the reactor volume, leading to higher concentrations of species B. Increasing the volumetric flow rate (v0) reduces the residence time in the reactor, meaning lower conversion of A and lower B formation.

2) Now, the reversible reaction. The following graphs have K as 0.5 and 10 respectively.





For a small equilibrium constant, the reaction strongly favors reactants, shown by the limited formation of B. As K increases, the equilibrium shifts toward products meaning higher conversion of A and higher B concentration. Increasing k accelerates the approach to equilibrium while increasing v0 reduces the extent of reaction.

3) Q1- How would volumetric flow that is not constant affect the concentrations of A and B? Q2- How can you find what reactor volume is required for different values of K?

4) When the reaction is reversible, the concentration of A doesn't approach zero like the irreversible case. CA approaches an equilibrium value (K). Because of this, the reversible reaction has lower conversion of A compared to irreversible.

P1-5) A(or 1) is correct. This is because A has the correct structure for the A balance and A disappears (negative reaction). The B balance is also correct because it has the same disappearance rate as A with the correct inlet/outlet species. There is no C in the feed and C is formed at twice the A consumption rate. All of the balances are correct.

The rest of the choices are not correct. B(2) has wrong stoichiometry. C(3) has wrong signs and rates. And D(4) swaps the A and B feed terms.